SAME IMAGE, DIFFERENT COVERAGE: AUDITING BATCH-ADAPTIVE MEDICAL CLASSIFIERS

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ABSTRACT

Prediction-time batch normalization makes a medical classifier's output depend on its batch companions. We investigate whether this dependence changes prediction-set coverage for an unchanged query image. A controlled audit evaluates batch-normalized architectures and a group-normalized control on three medical imaging benchmarks, using five training seeds and a fixed menu of companion contexts. A 224-pixel pretrained ResNet18 extension tests a higher-resolution setting. We distinguish average context coverage from true-label inclusion under every context. On PneumoniaMNIST, context-specific calibration of the compact residual model gives 92.9% mean coverage but 85.5% simultaneous coverage. We compare the coverage–size trade-offs of established simultaneous conformal corrections, including Corrected Max-Rank and maximum-score calibration. Duplicate checks, lesion-disjoint dermatology partitions, paired analyses, and calibration-split sensitivity support reproducibility. The findings motivate specifying companion-context assumptions when reporting calibrated medical-image predictions and provide an operational audit for inference scheduling.

***Index Terms*—** medical imaging, batch normalization, conformal prediction, deployment audit

1. INTRODUCTION

A hospital inference service may form a batch from the studies currently in a queue. With prediction-time batch normalization (BN), changing the other studies changes a query's normalized features even if its pixels remain identical. Prediction-time BN can improve robustness under covariate shift [1], but class imbalance and problematic companion samples are established failure modes [2, 3]. For a system returning a set of plausible diagnoses, a separate question arises: does its true-label inclusion claim survive alternative companion contexts?

Standard split conformal prediction protects an exchangeable future example for a specified scoring procedure [4]. It does not equate average coverage across inference contexts with inclusion for the *same* example under all contexts. This distinction matters when a service advertises a coverage level while its scheduler may change batch membership. We study the distinction by holding model weights and query pixels fixed and replacing only companions.

Our contributions are (i) an operational audit separating paired class changes, mean context coverage, and simultaneous context coverage; (ii) a comparison of ordinary, context-specific, and established simultaneous conformal corrections under identical predictions; and (iii) a reproducible study with duplicate checks, lesion-disjoint dermatology partitions, five training seeds, normalization controls, and calibration-split sensitivity. The contribution is a controlled empirical audit of an operational coverage claim; the underlying normalization and conformal methods are established. Code, weights, and per-image outputs are released.[1]

2. RELATED WORK

BN [5] couples examples through estimated moments. Prediction-time BN [1], TTN [6], DELTA [2], and MedBN [3] address adaptation, mixed moments, imbalance, and adversarial companions. Our intervention uses unchanged, unoptimized training images and freezes all learned parameters; it is not a reproduction of entropy-minimizing or continually updated adaptation.

Robust conformal prediction already calibrates restrictive scores over perturbations [7]. Max-Rank [8] exploits dependence for simultaneous control; we implement its finite-sample-corrected score-space version [9]. Batch conformal classification [10] concerns multiple queried examples, whereas our joint event concerns one query under several contexts. Medical VLM adaptation with conformal prediction is also established [11, 12]. Event reordering can also alter adaptive ECG conformal thresholds without changing signals or classifier outputs [13]. Here thresholds are stateless after calibration, and companion replacement changes classifier scores through BN. Our focus is the resulting all-context coverage estimand.

3. COMPANION-CONTEXT AUDIT

3.1. Fixed intervention and estimands

Let $f_\theta(x; B_j)$ be a trained K-class classifier evaluated on query x with $b-1$ fixed companions B_j. Let $\mathcal{B} = \{B_1, \ldots, B_M\}$ be chosen from training data independently of calibration and evaluation examples. For each query, normalization uses the batch $\{x\} \cup B_j$. Query groups are isolated even during vectorized inference. Each context therefore defines a fixed score

$$s_j(x, y) = 1 - [f_\theta(x; B_j)]_y. \tag{1}$$

For thresholds t_j, write $C_j(x) = \{y : s_j(x, y) \le t_j\}$. We report

$$\widehat{\mathrm{Cov}}_{\mathrm{mean}} = \frac{1}{NM} \sum_{i=1}^{N} \sum_{j=1}^{M} \mathbf{1}\{y_i \in C_j(x_i)\}, \tag{2}$$

$$\widehat{\mathrm{Cov}}_{\mathrm{all}} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\{y_i \in \cap_j C_j(x_i)\}. \tag{3}$$

The second event requires true-label inclusion under every menu context. Its failures can affect different images across contexts

[1] https://github.com/shi1720/medical-batch-coverage-audit

even if each context is individually calibrated. Per-context marginal coverage $1-\alpha$ alone yields only the union-bound lower bound $\max(0, 1-M\alpha)$ for the joint event, not $1-\alpha$. This is an elementary multiple-comparisons distinction, not a new theorem.

3.2. Calibration comparators and guarantee scope

For calibration scores $a_1, \ldots, a_n$, let $Q_\alpha(a)$ be the $\lceil (n+1)(1-\alpha) \rceil$th ascending value, using $+\infty$ when the rank exceeds n. We compare: *nominal*, which calibrates s_1 under the first natural context and reuses that threshold; *context-specific*, $t_j = Q_\alpha(s_j)$; *Bonferroni*, $t_j = Q_{\alpha/M}(s_j)$; *Corrected Max-Rank* (CMR) [9]; and a *score envelope*, $t_j = q$ for every j, where

$$S(x,y) = \max_j s_j(x,y), \qquad q = Q_\alpha\big(\{S(x_i,y_i)\}_{i=1}^n\big). \tag{4}$$

For CMR, compute maximum-tie ranks within each context, take $r = Q_\alpha(\{\max_j R_{ji}\}_i)$, and set t_j to context j's $(r+1)$st ascending calibration score, or $+\infty$. The additional order statistic follows [9]. Maximum-tie ranks and closed thresholds give a conservative tie extension, detailed in the reproducibility notes.

For a fixed model and menu, if calibration examples and the future query are exchangeable, the envelope satisfies

$$\Pr\{Y \in C_j(X) \text{ for all } j\} \geq 1-\alpha. \tag{5}$$

Indeed, the simultaneous event equals $S(X,Y) \leq q$; the usual exchangeable-rank argument applies to this scalar score. This is a direct application of robust split conformal prediction [7]. The probability is marginal over calibration and query, conditional on the fixed model/menu, not conditional on a particular realized calibration set. It excludes unseen contexts, changing model state, and distribution shift. Each deployed C_j needs one context pass; constructing the companion-independent intersection needs all M passes. Mean deployed set size is the primary efficiency metric. Empty sets are allowed; empty/singleton rates and intersection sizes are released.

4. EXPERIMENTS

4.1. Data integrity and protocol

We use 28-pixel PneumoniaMNIST, BreastMNIST [14, 15, 16], and the published DermaMNIST-C correction [17, 18]. Tasks are chest-radiograph pneumonia classification, ultrasound malignancy classification (normal/benign merged), and seven-class dermoscopic lesion classification. A label-blind SHA-256 audit removes exact image-byte duplicates with training, then validation, then test priority. Original PneumoniaMNIST has eight distinct train–validation overlaps; BreastMNIST has one train–test overlap. The correction's dermatology partitions are lesion-disjoint, verified using metadata. One image per test lesion is retained in metadata order before calibration splitting. Table 1 gives final counts. Patient identifiers are unavailable for these experiments; lesion separation and exact-byte removal do not establish patient independence.

Primary models are a three-block CNN-BN (16/32/64 channels), a compact residual BN model (16/32/64 channels), and an otherwise matched CNN with four-group GroupNorm [19]. We train five seeds, AdamW (learning rate 10^{-3}, weight decay 10^{-4}), batch 64, for 25 epochs without augmentation; the lowest validation negative log likelihood selects a checkpoint. Inputs are $x/127.5-1$. All training uses two CPU threads; vectorized inference uses Apple MPS, checked against CPU. We use population variance for current-batch inference, matching PyTorch training-mode BN with frozen affine parameters and no running-state updates.

Table 1. Final disjoint image counts; dermatology test units are distinct lesions. Calibration/evaluation halves use seed 20406.

Dataset	Train	Val.	Cal.	Eval.
PneumoniaMNIST	4690	516	310	311
BreastMNIST	546	78	77	78
DermaMNIST-C	8213	573	574	575

A secondary 224-pixel PneumoniaMNIST extension fine-tunes ImageNet-pretrained ResNet18 [20], all parameters, for five epochs on MPS (five seeds, AdamW 10^{-4} learning rate and weight decay, batch 32, no augmentation, lowest validation NLL). Grayscale inputs are repeated across RGB and ImageNet-normalized. Source-image indices follow primary cleaning, with an additional 224-pixel byte audit. BN replacement preserves pretrained state. RN18-224 uses the same context design and conformal methods.

The batch contains one query and 15 companions. For each of three fixed draws (seeds 801–803), the menu contains a natural-frequency context, an approximately class-balanced context, and one homogeneous context per class: $M = 12$ for binary datasets and $M = 27$ for dermatology. Homogeneous contexts deliberately stress composition; labels are used to construct the audit menu, not supplied to inference or claimed available to a scheduler. Balanced draws are independent within each class and can repeat images; exact context indices are released. The same menu and unstratified calibration/evaluation split apply across models and seeds. Random partitioning supports image-level finite-pool analysis, without establishing exchangeability of future patients. We set $\alpha = 0.1$ throughout.

The exploratory protocol and amendments are released. Exact-byte cleaning and the published dermatology correction preceded final training on the corresponding datasets; earlier pilots are excluded. Runtime motivated restricting mixed-moment, first-layer-only, matched affine-intensity, and batch-size controls to seed 0. CMR was added after the literature review. These choices are disclosed rather than described as preregistered. APS score sensitivity was added after the binary-dataset results. Repeated calibration splitting uses 100 fixed random splits of the same held-out pool; it measures split sensitivity, not 100 independent datasets.

4.2. Results

Table 2 reports all primary BN configurations and the pretrained RN18-224 extension. The source-statistics and GroupNorm controls are exactly companion-invariant in saved probabilities. Under current-batch BN, changing companions produces 13.1–51.0% class-flip rates across the six primary dataset–architecture settings. This is a paired intervention: the query itself is never perturbed.

The compact residual PneumoniaMNIST model illustrates why mean coverage is insufficient. Its context-specific mean coverage is 92.9%, but simultaneous coverage is 85.5%. The paired difference is 7.5 [5.9, 9.2] percentage points (95% query-bootstrap interval, conditional on the five fitted models and their calibration sets). CMR increases simultaneous coverage to 96.6%, with mean deployed set size 1.68 versus 1.28 for context-specific calibration. The envelope gives 96.1% with size 1.90. Thus the experiment does not establish the envelope as the most efficient correction; dependence-aware thresholds can be preferable.

Table 2. Current-batch BN: simultaneous coverage (%) / mean deployed set size. Each cell is the mean across five training seeds. Per-context calibration targets each context separately; simultaneous methods target the joint event. Full seed SDs, accuracy/AUC, and per-image outputs are released.

Dataset	Model	Flip (%)	Nominal	Per-context	Bonferroni	CMR	Envelope
PneumoniaMNIST	CNN-BN	16.2	85.1 / 1.19	90.9 / 1.19	99.4 / 1.77	92.5 / 1.23	92.2 / 1.33
PneumoniaMNIST	ResNet-BN	41.4	63.5 / 1.16	85.5 / 1.28	99.3 / 1.73	96.6 / 1.68	96.1 / 1.90
BreastMNIST	CNN-BN	13.1	83.6 / 1.14	85.9 / 1.13	100.0 / 2.00	91.5 / 1.26	92.8 / 1.48
BreastMNIST	ResNet-BN	51.0	57.4 / 1.14	77.7 / 1.24	100.0 / 2.00	92.1 / 1.59	92.6 / 1.82
DermaMNIST-C	CNN-BN	23.1	78.2 / 1.13	81.6 / 1.26	98.3 / 2.87	89.7 / 1.67	90.6 / 1.85
DermaMNIST-C	ResNet-BN	28.2	73.7 / 1.12	77.0 / 1.36	98.4 / 3.07	90.4 / 2.00	89.7 / 2.36
PneumoniaMNIST	RN18-224	44.8	55.5 / 1.01	83.0 / 1.17	98.1 / 1.63	93.6 / 1.37	92.0 / 1.74

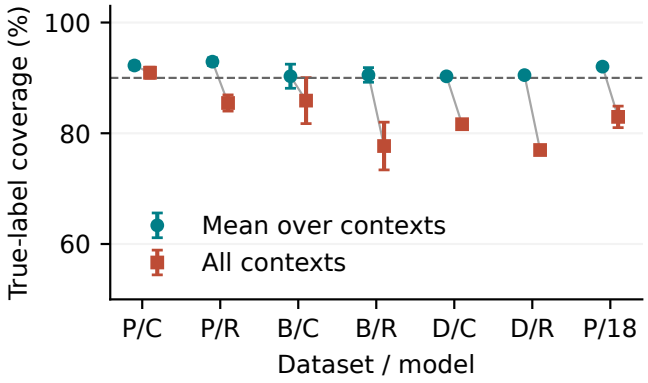


Fig. 1. Mean context coverage and simultaneous coverage under context-specific calibration. Points average five training seeds; bars show seed SD. P/B/D denote pneumonia/breast/dermatology; C/R denote compact CNN/residual models; 18 is pretrained RN18-224. The dashed line is the 90% target. Context-specific calibration does not target the joint event.

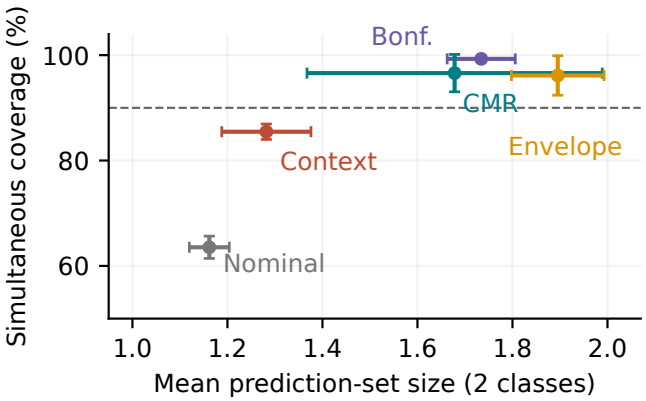


Fig. 2. Coverage–size trade-off for PneumoniaMNIST compact residual models. Points are five-seed means, bars are seed SD; size averages over queries and menu contexts. The envelope's common threshold can be conservative relative to CMR.

The 224-pixel pretrained RN18 extension has 91.0% source-statistics accuracy. Current-batch BN gives a 44.8% companion-induced class-flip rate; context-specific calibration yields 92.0% mean and 83.0% simultaneous coverage. CMR gives 93.6% simultaneous coverage with mean set size 1.37. This extension uses its own predefined fine-tuning protocol and does not establish clinical generalization.

Seed-0 normalization controls on PneumoniaMNIST yield class-flip rates of CNN-BN: 19.6% full BN, 15.4% mixed moments, and 15.1% first-layer-only; ResNet-BN: 45.3% full BN, 34.4% mixed moments, and 12.5% first-layer-only. The fixed 50:50 moment mixture is a control, not a full TTN implementation. Complete secondary results are released, including a matched affine-intensity condition and batch sizes 8 and 32.

An exploratory randomized APS sensitivity [21] shares one uniform variate across each query's contexts: the PneumoniaMNIST residual model gives 83.7% simultaneous coverage with context-specific calibration and 92.3% with CMR (size 1.37). Deterministic APS makes CMR and the envelope return both labels on this task. Score choice therefore changes the coverage–size trade-off; all score diagnostics are released.

A seed-0 validation-companion sensitivity gives 92.4% mean versus 86.2% simultaneous coverage for the compact pneumonia residual model. These companions were used for checkpoint selection but excluded from gradient training; results for all six configurations are released.

Finite calibration is visible, particularly on BreastMNIST. With only 77 calibration examples, Bonferroni's requested order statistic exceeds the available sample, returning the full label set. This is the correct finite-sample convention, not a numerical failure. Realized evaluation coverage can fall below 90% even for valid marginal methods; Fig. 3 separates this variation from a universal guarantee for each calibration split.

5. DISCUSSION AND LIMITATIONS

A calibrated service should specify whether its guarantee concerns a random realized context, each fixed context separately, or all allowed contexts for the same query. The latter can accommodate a context choice made after seeing the query, but only within the fixed, independently chosen menu. Our menu is a reproducible stress test, not an exhaustive hospital scheduler model. Protection against arbitrary companions does not follow.

A practical application is quality assurance for imaging inference services: replay unchanged studies under allowed batch compositions, measure paired prediction-set changes, and test whether reported coverage matches its stated estimand. Commercial utility, clinical benefit, patentability, and regulatory readiness remain untested.

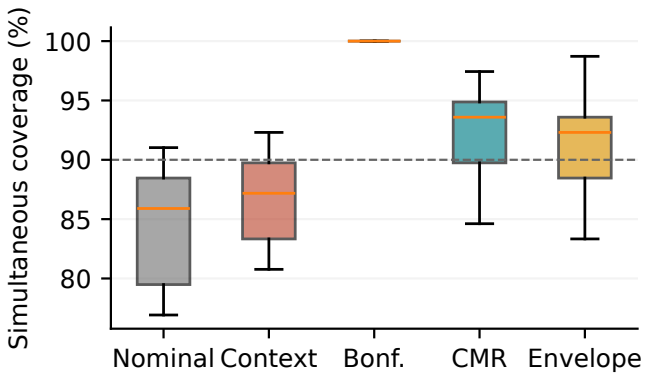


Fig. 3. Simultaneous coverage across 100 calibration/evaluation repartitions of the fixed BreastMNIST test pool, CNN-BN seed 0. Boxes show median and quartiles; whiskers show the 5th/95th percentiles. Repartitions are dependent sensitivity trials.

Most experiments use compact 28-pixel models; the 224-pixel extension uses one ImageNet-pretrained backbone. Diagnostic-resolution images and medical foundation models require separate evaluation. Missing patient identifiers prevent patient-level independence claims; dermatology metadata establish lesion separation only. Global coverage can conceal rare-class failures, and benchmark labels are not independently adjudicated diagnoses. Five seeds quantify training variability, not population diversity. The bootstrap conditions on fitted models and calibration. The exploratory protocol, small breast evaluation set, and hand-designed menu limit generalization. Patient-disjoint, multisite studies and gradient-based adaptation remain future work. Our conclusion is limited to measurable companion-induced differences between ordinary and simultaneous coverage.

6. COMPLIANCE WITH ETHICAL STANDARDS

This computational benchmark reuses public, de-identified image releases under their stated licenses: CC BY 4.0 for PneumoniaMNIST and BreastMNIST, and CC BY-NC 4.0 for DermaMNIST-C/HAM10000-derived images. No source images are redistributed with the software. The dermatology experiments do not establish permission for commercial use of the underlying data. Predictions are research outputs and are not offered as clinical decisions.

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